

ICT Project (Individual)

HNDIT1052– Individual Project

**QR-Based Job Record and Management System
RAT/IT/2023/F/0017**

Department of Information Technology

Advanced Technological Institute

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Introduction

Name Of the Client:

Deshapriya Motors
No. 105/10,
Panadura Road,
Ratnapura [1]



Image 1

Deshapriya Motors is an authorized dealership of David Pieris Motor Company (Pvt) Ltd, located in the Ratnapura area near Saman Dewalaya on Panadura Road. It provides automobile service and maintenance for motorcycles, three-wheelers, and Bajaj Qute four-wheelers.

Currently, the owner of the service center manages daily jobs and sales using written records, which has been effective so far. However, with the business growing and the number of employees and daily jobs increasing, the owner feels that a paper-based system has reached its limit.

While the parent company already provides a system for maintaining sales records for accounting purposes, the client now needs a system to track the number, type, and time range of jobs. This system should also be able to calculate employee incentives.

Background and Motivation

Daily reports play a crucial role in employee time management and productivity. They help employees become more accountable, enhance their productivity, and provide a structured way to communicate with their supervisors. It also contributes to better project tracking and overall time-saving. [2]

As a growing entrepreneur at a start to mid-level, the client is constantly looking for opportunities to expand his business. However, given the current economic situation in Sri Lanka, this is a challenging task without significant capital investment. This project aims to be a solution by providing support with minimal resource requirements, utilizing existing resources and capital.

While the current system is functioning to some extent, it is far from being optimal or efficient. The paper-based system, heavily dependent on manual input from the owner or management to calculate incentives, is neither motivating nor inspiring from an employee perspective. Furthermore, managing wages and incentives through verbal or paper-based communication adds to the owner's workload, which is already considerable as he also works as a technician in his shop.

Since the dealership also serves as an authorized distributor of spare parts, it is crucial for our client to stay ahead of current trends in the repair industry. This requires keeping track of the most common repairs and ensuring the necessary parts are in stock for swift and efficient service, encouraging customers to return.

As someone who is planning to complete a Higher National Diploma in Information Technology, I am confident in my ability to offer a simple, low-cost technological solution that can operate with the existing resources. This solution can support the business's growth until it is ready to invest in a more professional-grade system.

Goal, Objectives and Scope of the project

The primary goal of this project is to develop a QR-based job record system that improves the efficiency and accuracy of job tracking, employee performance management, and inventory control for Deshapriya Motors. The system should streamline the process of recording and managing service jobs, automate employee incentive calculations, and optimize inventory management, all while utilizing minimal resources and leveraging existing infrastructure.

We hope to operate within the following stated scope on this project.

1. Development of a QR-Based Job Tracking System:

Design and implement a digital system that uses QR codes to track jobs for motorcycles, three-wheelers, and Bajaj Qute four-wheelers.

Generate unique QR codes for each job or service task, which can be scanned to log job details, start times, and the employee assigned.

Enable the system to automatically record job data, reducing the reliance on manual entry and paper-based records.

2. Employee Performance and Incentive Management:

Develop a feature that automatically calculates employee incentives based on completed jobs, job complexity, and efficiency.

Provide real-time updates on employee performance, allowing management to monitor and reward employees more effectively.

Include functionalities for employees to view their performance metrics and incentives.

3. User-Friendly Interface and Accessibility:

Develop a user-friendly interface that can be easily accessed by management and employees on various devices, such as smartphones, tablets, or computers.

Ensure that the system is intuitive and requires minimal training for users to operate effectively.

4. Testing and Deployment:

Conduct thorough testing of the system to identify and resolve any issues before full deployment.

Provide support during the initial deployment phase to ensure a smooth transition from the existing paper-based system.

5. Training and Support:

Offer comprehensive training for all employees and management on how to use the new system effectively.

Expected Outcomes:

- Improved operational efficiency through automated job tracking and incentive management.
- Enhanced accuracy in recording job details and managing employee performance.
- Optimized inventory control, reducing the risk of stockouts and overstocking.
- Increased employee motivation through transparent and timely incentive calculations.
- Scalability for future business growth and system enhancements.

Proposed Solution

When you first create systems, it is best to keep them as simple and straightforward as possible. But as those systems begin to get developed, they will naturally allow for more complexity. [3]

QR Code Job Tracking and Management:

- **Description:** This functionality allows for the generation and scanning of unique QR codes for each service job. When a job is initiated, a QR code is created and attached to it. Employees can scan this QR code to log job details, including start time, end time, type of service performed, and employee assigned.
- **Benefits:** This feature streamlines the recording process by reducing manual data entry, minimizes errors, and provides real-time updates on the status of each job. It improves workflow management by offering clear visibility into ongoing tasks and job progress.

Automated Employee Performance and Incentive Calculation:

- **Description:** This function automatically calculates employee incentives based on predefined criteria, such as the number of jobs completed, job difficulty, and efficiency. The system tracks each employee's performance metrics using the data collected from the QR code scans.
- **Benefits:** By automating incentive calculations, the system ensures a fair and transparent process, motivates employees through timely rewards, and reduces the administrative workload for management. This leads to increased employee satisfaction and productivity.

Reporting and Analytics:

- **Description:** The system provides comprehensive reporting and analytics capabilities. It can generate various reports, such as job completion rates, employee performance summaries, and service trends. The data is visualized in an easy-to-understand format to aid decision-making.
- **Benefits:** This feature enables management to make data-driven decisions,

identify areas for improvement, and plan for future needs. It also helps in monitoring business performance and understanding customer demand patterns.

These critical functionalities—QR Code Job Tracking and Management, Automated Employee Performance and Incentive Calculation, and Reporting and Analytics—are designed to enhance the operational efficiency of Deshapriya Motors. They aim to reduce manual workload, increase accuracy, improve employee motivation, and provide valuable insights into business performance, contributing to overall business growth and success.

Table 1

ISSUE NO.	STATUS	PRIORITY	CATEGORY	ASSIGNED TO	DATE OPENED	COMMENTS
	Complete	High				
	Complete	High				
	In Progress	High				
	Overdue	High				
	Not Started	Medium				
	On Hold	Low				

Table 2

STATUS KEY	PRIORITY KEY
Not Started	High
In Progress	Medium
Complete	Low
On Hold	
Overdue	

Resource Requirements

To undertake the development of a QR-based job record system for Deshapriya Motors, we will need various resources to ensure the project is completed efficiently and effectively. Here's a breakdown of the resource requirements for a simple implementation of the project.

Hardware:

- **Smartphones and/or Tablets:** Depending on the number of employees, it is recommended to allow the use of personal devices when possible and provide company devices only when necessary.
- **Computer and/or Laptop:** For this project, a mid-level computer is required for development tasks. Additionally, a computing device should be available at the shop to manage the backend of the implemented system.

Software:

- **QR Code Generation and Management Software:** Software needed to create and manage QR codes for each job or service task.
- **Job Record and Management System:** Core software used for tracking jobs, logging details, and calculating employee incentives.
- **Database Management System:** A system for storing all job data, employee information, and performance metrics.

Networking and Infrastructure:

- **Internet Connection:** A reliable internet connection is essential for connecting all devices to the central system and ensuring data synchronization.

Project Plan

Task	Duration	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13
Define Project Scope	1 week													
Resource Allocation	1 week													
Gather Requirements	1 week													
Develop Project Plan	1 week													
Design System Architecture	1 week													
QR Code System Design	1 week													
Job Record System Design	1 week													
Develop QR Code Module	1 week													
Develop Job Record Module	2 weeks													
Set Up Database	1 week													
Integrate Systems	1 week													
System Testing	2 weeks													
Prepare Training Materials	1 week													
Test Deployment	1 week													
Launch System	1 week													
Provide Support	2 weeks													
Final Documentation & Review	1 week													

References

- [1] D. P. M. C. (. Ltd, "Deshapriya Motors Dealer," David Pieris Motor Company (Pvt) Ltd, [Online]. Available: <https://www.dpmco.com/en/dealer/deshapriya-motors.html>.
- [2] T. Shetty, "Why Is Daily Activity Report Important For Employee Time Management?," EmpMonitor, 2023. [Online]. Available: <https://empmonitor.com/blog/daily-activity-report-importance-in-workplace/>.
- [3] S. Miker, "Simple systems are better than optimized systems to start," 2019. [Online]. Available: <https://www.scottmiker.com/improving-systems-and-habits/2019/6/11/simple-systems-are-better-than-optimized-systems-to-start>.

